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K-Ar ages and magnetic stratigraphy of a hotspot-induced, fast grown oceanic island: El Hierro, Canary Islands

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Abstract

The combined use of accurate radiometric dating and magnetic stratigraphy can be applied to define the main stages of the building of oceanic volcanic islands. This method has been successfully applied on the island of El Hierro, the westernmost island of the Canaries Archipelago. For the emerged part of this island, built in the last 1.2 Ma, magnetic stratigraphy and mapping show the presence of four consecutive subchrons. K-Ar dating of eighteen samples in stratigraphic sequences using an unspiked technique produced ages consistent with the general volcanic stratigraphy. The ages show that the magnetic polarity zones correlate to the upper part of the Matuyama and to the Brunhes chrons. The Jaramillo normal polarity subchron of the upper part of the Matuyama reverse polarity Chron is recorded in one of the lava sequences. The lower limit of this event is constrained by two ages of 1.04 ± 0.02 Ma and one of 1.05 ± 0.02 Ma, close to the value deduced from the astronomically tuned magnetic polarity time scale (APTS).

The volcanic history can be summarised by two consecutive main basaltic volcanic edifices: the El Tinor, active during the upper part of the Matuyama Chron and the El Golfo edifice, constructed during the Brunhes Chron.

Keywords: El Hierro, Canary Islands; K-Ar geochronology; paleomagnetism; volcano magnetic-stratigraphy; Jamarillo subchron

1. Introduction

Volcanic islands are formed by a competition between constructive and destructive processes. The precise reconstruction of the volcanic history of El Hierro may be a key factor in the understanding of the origin and development of structural features common to Canarian and other volcanic islands such as three-armed rift structures (Wentworth and Macdonald, 1953; Walker, 1992; Carracedo, 1994) and giant gravitational collapses (Lipman et al., 1988; Holcomb et al., 1988; Moore et al., 1989; Masson, 1994).

The published K-Ar ages are notably inconsistent, ranging from 3.05 ± 3 to 0.19 ± 0.01 Ma (Abdelmonem et al., 1972) or from 0.80 ± 0.06 to less than 0.05 Ma (Fúster et al., 1993). Reverse polarity volcanic units which have not been previously identified were found in a preliminary paleomagnetic study of

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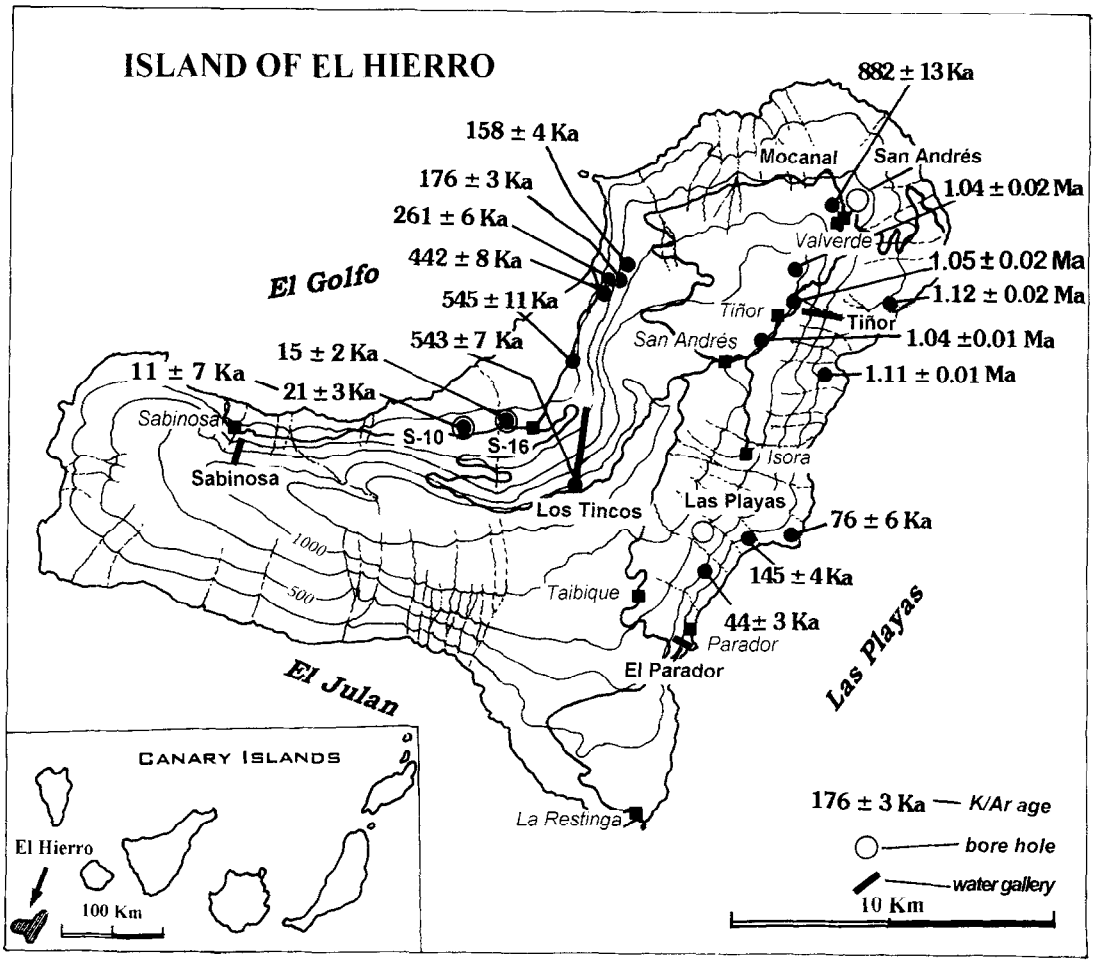
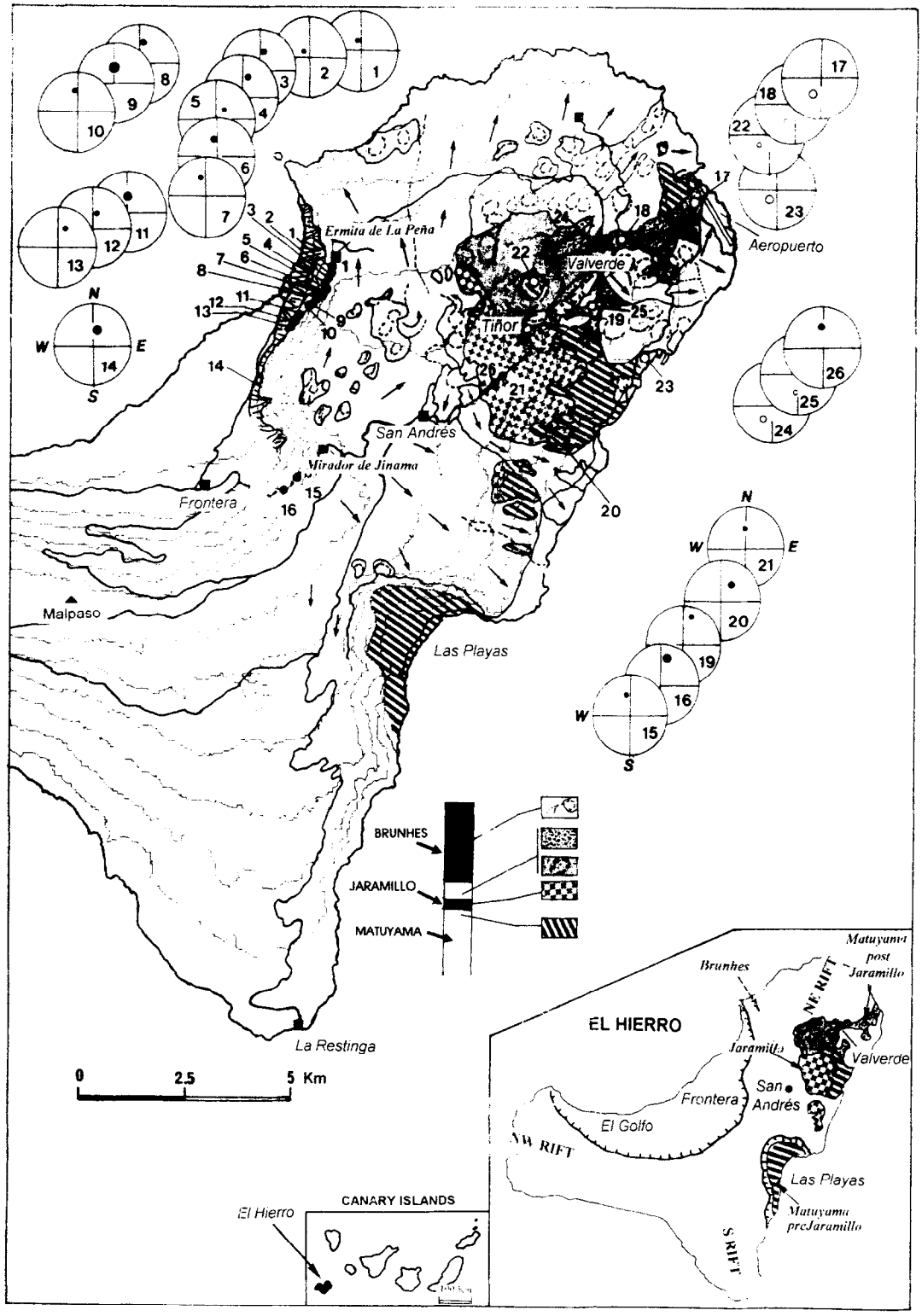


Fig. 1. Map of El Hierro showing location of samples dated (K/Ar method). The boreholes and water galleries locations are also shown. Notice that some of the ages were obtained from samples inside boreholes.

the lavas of El Hierro (Carracedo, 1994). The presence of reverse polarity units is not consistent with the assumption that the island evolved during the normal polarity Brunhes epoch (Fúster et al., 1993).

On the basis of the existing geological map of the island (Pellicer, 1977), a new sampling campaign has been carried out on the main volcanic units for both K-Ar and paleomagnetic measurements. Only the eastern part of the island, which has not been entirely covered by the most recent volcanic phase, has been investigated. Two volcanoes, the El Tinor and the El Golfo edifices and recent volcanic bodies related to the present-day northeastern rift are located in this area. Magnetic polarity stratigraphy has been applied to the vocanic units. We have correlated these magnetozones to the volcanic units. When possible, we have selected samples which separate each volcanic unit to estimate the duration of the volcanic phases using K-Ar dating. Since the most recent series are below the limits of detection by conventional K-Ar

Fig. 2. Map showing the extension of the magnetozones detected on El Hierro. The location and mean directions of the remanent magnetisation of sites shown in Table 1 are also indicated. The size of the circles indicates 95% confident level. Solid = normal polarity; open = reverse polarity.


Ermita de La Peña
Valverde
Tiñor
Aeropuerto
San Andrés
Mirador de Jinama
Frontera
Malpaso
Las Playas
La Restinga
BRUNHES
JARAMILLO
MATUYAMA
0 2.5 5 Km
EL HIERRO
Brunhes
Matuyama post Jaramillo
NE RIFT
Jaramillo
Valverde
San Andrés
Frontera
El Golfo
NW RIFT
Las Playas
Matuyama preJaramillo
S RIFT
CANARY ISLANDS
El Hierro

dating methods (Fúster et al., 1993), Ar isotopic measurements have been done using an unspiked technique (Cassignol et al., 1978) developed specifically to date recent rocks (i.e., ages lower than 0.1 Ma).

This paper focuses mainly on the geochronology and magnetostratigraphy of El Hierro. The detailed reconstruction of the volcanic and structural evolution of the island, now in preparation, will be discussed elsewhere.

2. Geological background and previous work

The island of El Hierro is the emergent summit, some 280 km^2 in area, of a volcanic shield which rises from a 156-Ma-old seafloor at a depth of 3700–4000 m at the westernmost edge of the Canarian Archipelago. The island is believed to be built over the present location of the Canaries Hotspot (Holik et al., 1991; Hoernle et al., 1991).

The characteristic trilobate form of the island (Fig. 1 and inset in Fig. 2) is characteristic of three-armed rifts (Carracedo, 1994, 1995; Carracedo et al., 1995a). On El Hierro, the rifts form topographic ridges at angles of 120°, oriented in NW, NE and S directions. Unlike the island of La Palma, which had seven eruptions in historic time (the last 500 years), only one eruption (Lomo Negro volcano, 1973) occurred in El Hierro in this period (Hernández Pacheco, 1982). Nevertheless, this island shows the greatest concentration of recent, well-preserved emission vents in the Canaries Archipelago, with the vents tightly grouped along the afore-mentioned rifts. We have dated a lava from a group of recent emission vents at the central plateau, near the village of San Andrés (Mña. Chamuscada–Mña. Entremontañas volcanic group), with a ^{14}C age of 2500 ± 70 yr B.P. However, there seem to be many vents of similar or even younger age, particularly at the westernmost tip of the northwestern rift.

The breaches between the rifts form large, very conspicuous depressions: the El Julan (facing southwest), Las Playas (southeast) and El Golfo (north) embayments (see Fig. 1 and inset in Fig. 2). These large cliff-bounded basins have been related to major collapse structures (Ridley, 1971; Hausen, 1973; Bravo, 1982; Holcomb and Searle, 1991; Carracedo, 1994; Carracedo et al., 1995b). The vertical cliffs of the Las Playas (900 m) and El Golfo (1100 m) embayments show large sections unmasked by post-collapse lavas or vegetation, in which the main part of the stratigraphic sequences of the island can be observed. Valuable information about the deep structure and stratigraphy of the island has been obtained through galleries excavated for underground water mining and boreholes with core recovery drilled for water exploration (SPA-15, 1975).

The absence of reliable island-wide stratigraphic markers and the major block-tectonics complicate any precise reconstruction of the volcanic history of this island based solely on radiometric dating and standard geological observations. Mapping with K-Ar dated geomagnetic reversals have allowed needed stratigraphic correlation between sections, permitting the reconstruction of the geological evolution of this volcanic edifice.

3. Sampling and analytical techniques

The combined use of reversals and radiometric dating to define the main stratigraphic units and reconstruct volcanic histories, has proved to be efficient elsewhere in the Canary Islands (Watkins, 1973; Carracedo, 1979; Carracedo and Soler, 1995; Pérez Torrado et al., 1995). In these works, few, however, well-defined and correlatable magnetic reversals have been observed to occur in the islands of the Canaries Archipelago. Mapping and correlation of geomagnetic polarity units can afford a fast and easy procedure to "bracket" the duration of volcanic events determined by radiometric methods, greatly reducing the number of ages required.

3.1. Paleomagnetism

The preliminary definition of polarity units present on El Hierro was carried out in the field using portable fluxgate magnetometers.

A close correlation of polarities determined on the field with the stable laboratory directions after progressive thermal and alternating field demagnetisation has been observed (Carracedo, 1979; Carracedo

and Soler, 1995). Oriented samples (145 samples) were taken from 26 sites (see Fig. 2 and Table 1) of each of these polarity units for the analysis of the characteristics and stability of the remanent magnetisation. Each site consists of several cores taken several meters apart from a lava flow or contiguous lava flows of similar polarity. Standard procedures (portable drills and sun and magnetic compass orientation) were used. Two pilot standard cores from each site were prepared for paleomagnetic measurement. The direction and intensity of magnetisation were analysed using spinner magnetometers. Thermal and alternating field (AF) stepwise demagnetisation were carried out in the two pilot samples, respectively.

The main objective in the laboratory analysis was to confirm the stability of the remanent magnetisation of the samples to ensure the reliability of mea-

Table 1
Paleomagnetic data, El Hierro, Canary Islands

Site	K-Ar age	m a.s.l.	N	D (°)	I (°)	a_{95} (°)	K	Lat. (°)	Long. (°)
Post-El Golfo rift volcanism. Upper part of the Camino de La Peña (El Golfo cliff) (Brunhes)									
1		730	4	345.3	60.3	3.5	706	72.0	303.9
2		720	3	326.9	69.8	2.4	2556	54.6	308.0
3		700	3	8.5	42.6	4.8	660	81.8	92.5
4		680	4	16.1	37.2	6.0	236	73.7	94.1
5		670	5	30.8	60.5	2.2	1218	61.4	395.3
6	158 ± 4 ka	650	3	355.0	41.8	4.4	770	84.2	213.3
El Golfo volcanic edifice. Camino de La Peña (El Golfo cliff) (Brunhes)									
7	176 ± 3 ka	590	8	349.9	40.4	3.6	235	79.7	227
8		515	3	8.2	34.6	5.2	571	78.5	119.6
9	261 ± 6 ka	500	4	357.6	48.4	10.7	74	87.4	289.9
10		450	4	356.2	31.7	3.1	853	78.8	181.0
11	442 ± 8 ka	290	7	14.9	49.7	8.2	55	76.7	56.5
12		135	12	10.7	48.1	4.4	100	80.5	61.5
13		120	8	18.5	37.8	3.3	285	71.9	89.3
14	545 ± 11 ka	85	4	359.8	42.4	6.0	232	76.8	164.7
Camino Mirador de Jinama–Frontera (El Golfo cliff) (Brunhes)									
15		900	3	348.1	33.7	2.8	1891	75.7	214.0
16		770	4	12.4	23.2	7.1	167	70.5	123.1
Ventejís emission center. Upper part of Tiñor volcanic edifice (Matuyama–post-Jaramillo)									
17		100	5	195.6	−48.3	6.9	125	−76.3	242.2
18		500	8	196.9	−44.0	3.9	205	−74.8	255.6
24	882 ± 13 ka	640	7	210.8	−49.5	2.0	881	−63.0	239.2
25		710	7	186.1	−48.6	1.6	1350	−84.4	232.0
Ventejis emission center. El Tiñor plateau lavas (Matuyama–Jaramillo)									
19		820	6	15.9	14.9	2.4	785	64.0	123.7
20		170	7	33.3	40.3	3.4	324	59.6	73.6
21	1.04 ± 0.01 Ma	960	9	356.7	41.1	3.3	768	83.8	212.9
26		945	7	354.1	37.2	2.6	532	81.2	200.9
Early Tiñor steep dipping lavas (Matuyama Reverse Polarity Epoch–pre-Jaramillo)									
22	1.05 ± 0.02 Ma	920	4	202.2	−53.8	2.7	1140	−69.9	227.2
23	1.12 ± 0.02 Ma	80	6	214.9	−53.1	5.5	148	−59.5	232.3

N = number of samples at the site. Declination and inclination after stepwise demagntization. K-Ar ages refer to Table 2.

surements carried out in the field with portable magnetometers.

3.2. K-Ar dating

Once the magnetic properties of the samples were defined, eighteen samples were selected for K-Ar age determinations (see Fig. 1). The rocks have been collected almost entirely in the northeastern part of the island, where the four different magnetozones have been identified and mapped. When possible, samples were collected in stratigraphic sequence (boreholes and in the El Golfo escarpment) to verify the geological significance of the obtained results and also the accuracy of the dating technique, particularly in the youngest volcanic suites (i.e., El Golfo escarpment). The rest of the ages were obtained for the cross-correlation of the different magnetozones and to confirm the validity of the magnetostratigraphy previously defined. The youngest lavas sampled were from the rifts and from flows filling the El Golfo embayment, the younger lavas being taken from boreholes.

Only rocks without traces of alteration were selected. All the dated samples are characterised by loss on ignition (LOI) lower than 1.3%. The samples have been crushed and sieved to 0.25–0.125 mm size fraction and ultrasonically washed in HNO_3, in order to remove phases formed by alteration. Both K and Ar measurements were performed on the microcrystalline groundmass which is considered to be representative of the phase that has crystallised during the solidification of the lavas. Phenocrysts and xenocrysts, which may carry excess ^{40}Ar, were removed using heavy liquids of appropriate densities, and magnetic separations.

Potassium has been analysed by atomic absorption and flame emission spectrophotometry. Argon was extracted from 1 to 2.5 g groundmass samples by radio frequency heating induction in a high vacuum glass line, and purified with titanium sponge and Zr-Al getters. Argon was analysed using a 180°, 6 cm radius, 620 V accelerating potential mass spectrometer working in a semi-static mode. The isotopic composition of Ar and Ar content were determined using an unspiked K-Ar technique (Cassignol et al., 1978). Argon measurements rely upon the analysis of reference standard minerals. As deduced from the cross-calibration of GL-O (Odin, 1982), Mmhb-1 (Samson and Alexander, 1987), LP-6 (Odin, 1982) and HD-B1 (Fuhrmann et al., 1987) Ar contents were determined with a precision of 0.4% ($\pm 2\sigma$).

4. Results

4.1. Paleomagnetism

4.1.1. Reliability of the measurements

Two pilot samples of each site were stepwise thermally and in alternating field (AF) demagnetised in temperatures of 20 to 600°C or fields up to 600 mT (Fig. 3). No substantial change in directions is shown by the samples during the demagnetisation (see Fig. 3A). Viscous and stable components show consistently the same direction and none of the samples reversed polarity during the demagnetisation process. This corroborates the reliability of field polarity measurements and previous works on the Canary Islands (Carracedo, 1974, 1979) where geomagnetic polarities were measured directly in the field using portable fluxgate magnetometers.

4.1.2. Arrangement of the four magnetozones

Understanding of the island's internal structure was greatly improved by means of observations made in galleries (tunnels excavated for ground water mining) and cores (Fig. 1) from boreholes (SPA-15, 1975). The use of magnetic polarity mapping provided evidence of the presence on El Hierro of four distinct polarity zones or magnetozones (Fig. 2).

The lowest reverse magnetozone (R_1) appears only along the northeastern slopes of the island (Fig. 2), generally in lavas dipping steeply (30–35°) towards the sea from the northeastern ridge. The R_1 lavas were correspondingly detected in galleries and boreholes (Fig. 5).

Normal polarities corresponding to the N_1 magnetozone were detected in subhorizontal lavas overlying R_1 at the San Andrès plateau. Some of these lavas spilled over the central plateau and adapted to the steep dipping of R_1 lavas, apparently filling canyons carved in the R_1 volcanic suite (Fig. 2).

A few xenolith-rich lavas (a very distinct mapping marker since xenoliths are generally absent from the lavas of El Hierro) of reverse polarity (R_2) cap the

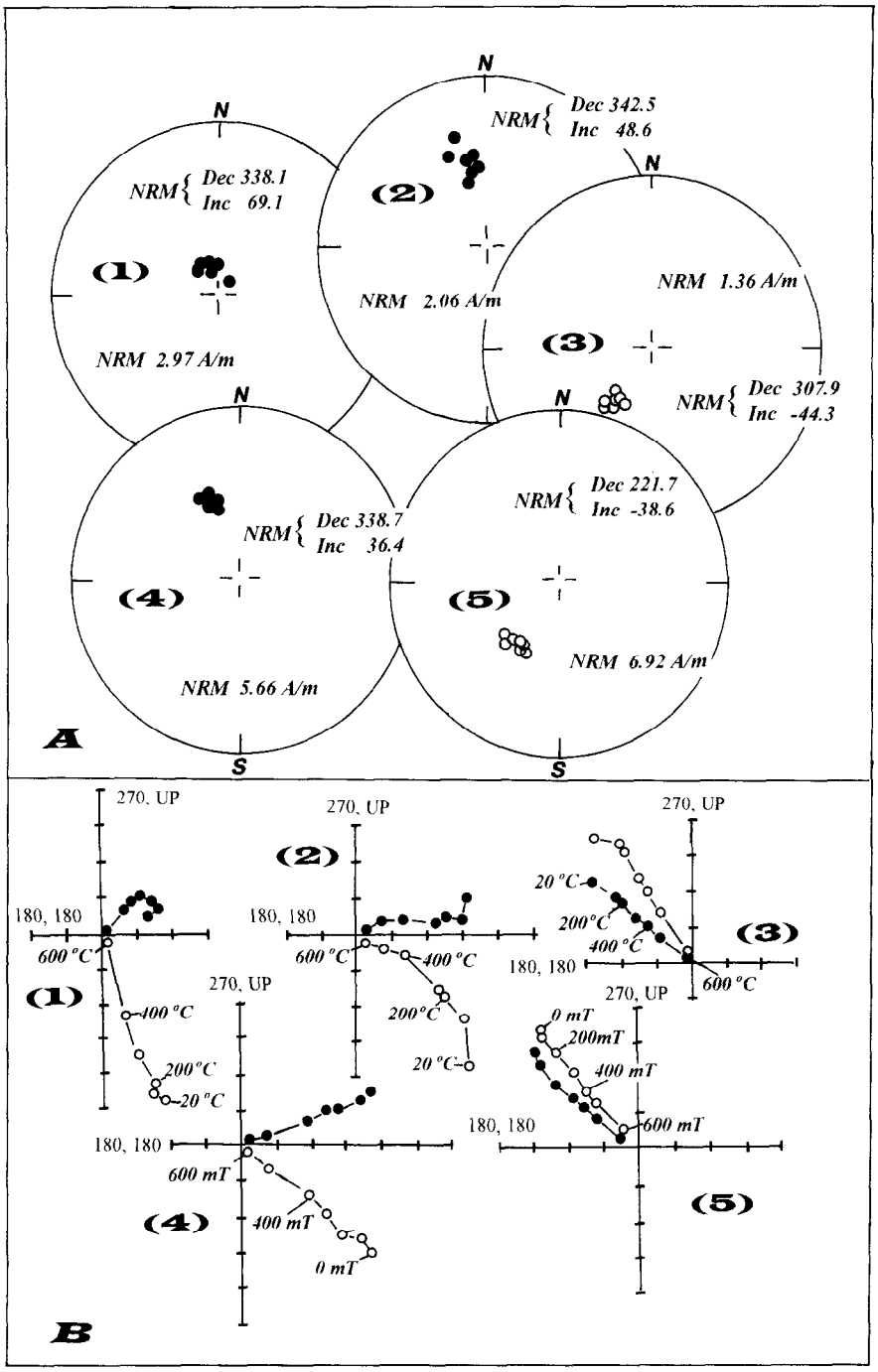


Fig. 3. (A) Changes in direction of the magnetisation vector during stepwise demagnetisation (20 to 600°C or 0 to 600 mT) of representative samples of the different magnetozones detected on El Hierro. No significant variations are detected, consistent with the assumption that polarities can be reliably determined in the field with portable flux-gate magnetometers. (B) Demagnetisation diagrams of the same samples. See explanation in the text. Samples *1*, *3*, *4* and *5* are basalts; sample *2* is an intermediate (benmoreite) rock.

aforementioned volcanic formations (see Fig. 2 and Fig. 5). The R_2 lavas seem to correspond to a late event related to emission vents of the Ventejis group, a cluster of cinder cones northwest of the town of Valverde; they flow downslope towards the sea over an already deeply eroded topography carved on the earlier volcanic formations.

Finally, lava flows and cinder cones of normal polarity unconformably overlie R_2 volcanics. No evidence of any internal discordance or erosional gap was observed in this apparently late-stage, rift-originated volcanism, grading at the top to emission vents of subhistoric age. We assign accordingly these recent volcanics to the Brunhes normal polarity chron. Magnetic polarity profiles carried out in the 1100 m vertical escarpment of El Golfo evidenced the presence only of normal polarity volcanics (Fig. 2). Normal lavas likewise fill this embayment, as evidenced by the 280 m deep S-16 borehole (Fig. 5). The galleries located at the foot of the El Golfo escarpment show consistently normal polarity lava flows and dykes (Fig. 4).

The limited outcrop of reverse polarity lavas, confined to the northeastern slopes, the presence of the Jaramillo normal lavas and the catastrophic disruption of the island by several giant landslides may explain the difficulties encountered in previous works for a correct reconstruction of the volcanic historic of this island.

4.2. K-Ar results

The location of the dated samples is shown in Fig. 1. Age determinations on the eighteen samples are presented in Table 2, which also shows the reproducibility of the results for independent replicate measurements. Standard errors (95% confidence level) were calculated from 100 measurements for each Argon isotope. The errors on the ages are calculated according to the equations given in Gillot

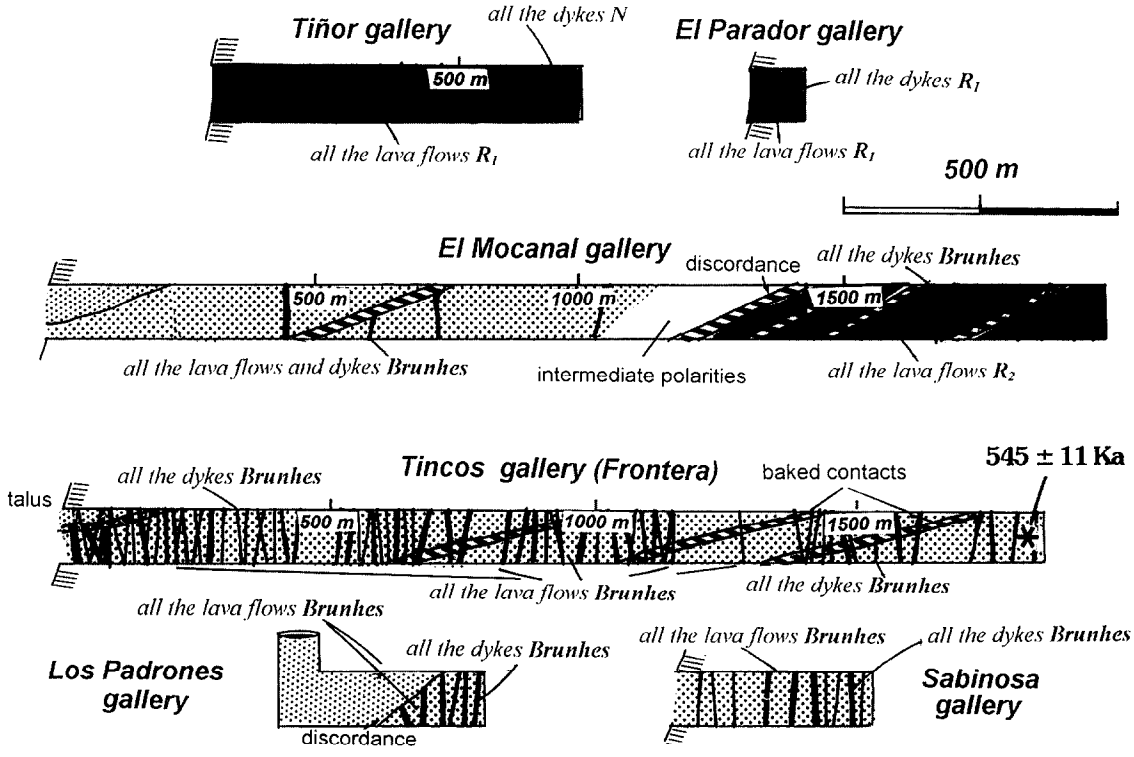


Fig. 4. Magnetozones detected in water galleries (2 m wide tunnels excavated for underground water mining) on El Hierro. See location of the gallerieFig. 4 and Fig. 1.

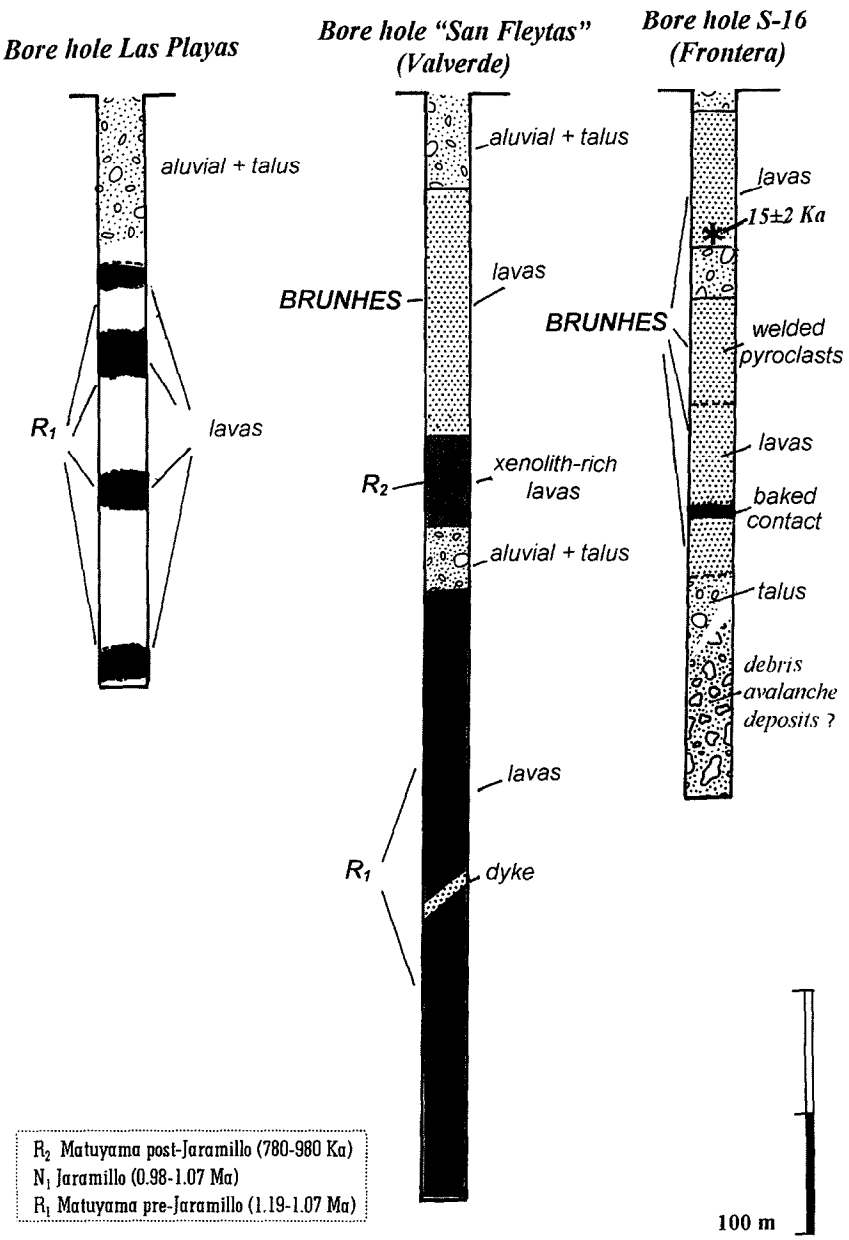


Fig. 5. Magnetozones found in borehole cores (see location in Fig. 1) on El Hierro.

et al. (1982). For the youngest dated samples (from 500 to 15 ka) the ages have a precision ranging between 14 and 2%. In this particular case, the precision is mainly related to the amounts of $^{40}Ar^{*}$ (radiogenic argon). For the older samples, the ages have a precision of about 1%. For the discussion we will use the mean value of the replicate measurements

The precise dating of R_1 magnetozone brackets the rest of the overlying magnetozones (N_1 and R_2) by correlation of chrons or subchrons to the GPTS. Only the normal polarity sequence (or sequences) at

Table 2
K-Ar Ages for volcanic rocks from El Hierro

Sample	P	K%	$^{40}Ar^{*}$ %	$^{40}Ar^{*}$ at $g^{-1} \times 10^{11}$	Age ($\pm 2\sigma$)	Age mean value
Rift volcanism						
CI HI 16_1	N	0.971 ± 0.010	0.162	0.125 ± 0.077	11 ± 8 ka	
CI HI 16_2	N	0.971 ± 0.010	0.209	0.125 ± 0.060	12 ± 6 ka	12 ± 7 ka
CI HI 11_1	N	1.571 ± 0.016	0.294	0.252 ± 0.040	15 ± 3 ka	
CI HI 11_2	N	1.571 ± 0.016	0.300	0.233 ± 0.037	14 ± 2 ka	15 ± 3 ka
CI HI 07_1	N	1.094 ± 0.011	0.455	0.243 ± 0.039	21 ± 3 ka	
CI HI 07_2	N	1.094 ± 0.011	0.404	0.243 ± 0.029	21 ± 3 ka	21 ± 3 ka
CI HI 15_1	N	1.383 ± 0.014	1.476	0.624 ± 0.042	44 ± 3 ka	
CI HI 15_2	N	1.383 ± 0.014	1.374	0.646 ± 0.047	44 ± 3 ka	44 ± 3 ka
CI HI 13_1	N	0.906 ± 0.009	1.777	0.742 ± 0.042	79 ± 5 ka	
CI HI 13_2	N	0.906 ± 0.009	0.596	0.693 ± 0.056	74 ± 6 ka	76 ± 6 ka
CI HI 14_1	N	1.731 ± 0.017	3.649	2.585 ± 0.072	143 ± 4 ka	
CI HI 14_2	N	1.731 ± 0.017		2.645 ± 0.071	147 ± 4 ka	145 ± 4 ka
CI HI 01_1	N	1.278 ± 0.013	3.629	2.036 ± 0.057	153 ± 5 ka	
CI HI 01_2	N	1.278 ± 0.013	4.237	2.174 ± 0.053	163 ± 4 ka	158 ± 4 ka
El Golfo volcanic edifice						
CI HI 02_1	N	1.942 ± 0.02	7.715	3.647 ± 0.051	180 ± 3 ka	
CI HI 02_2	N	1.942 ± 0.02	8.090	3.676 ± 0.046	171 ± 3 ka	176 ± 3 ka
CI HI 03_1	N	0.956 ± 0.009	4.491	2.525 ± 0.058	253 ± 6 ka	
CI HI 03_2	N	0.956 ± 0.009	4.587	2.684 ± 0.060	269 ± 7 ka	261 ± 6 ka
CI HI 04_1	N	0.903 ± 0.009	7.233	4.144 ± 0.061	439 ± 8 ka	
CI HI 04_2	N	0.903 ± 0.009	7.294	4.191 ± 0.061	444 ± 8 ka	442 ± 5 ka
CI HI 27_1	N	1.170 ± 0.012	11.093	6.562 ± 0.068	537 ± 8 ka	
CI HI 27_2	N	1.170 ± 0.012	15.526	6.696 ± 0.055	548 ± 7 ka	543 ± 8 ka
CI HI 05_1	N	0.986 ± 0.010	7.369	5.657 ± 0.082	549 ± 10 ka	
CI HI 05_2	N	0.986 ± 0.010	5.072	5.560 ± 0.113	540 ± 12 ka	545 ± 11 ka
El Tiñor volcanic edifice						
CI HI 06_1	R	0.722 ± 0.007	10.350	6.608 ± 0.072	876 ± 13 ka	
CI HI 06_2	R	0.722 ± 0.007	9.658	6.685 ± 0.077	886 ± 14 ka	882 ± 13 ka
CI HI 18_1	N	1.153 ± 0.012	17.886	12.426 ± 0.093	$1.03 + 0.01$ Ma	
CI HI 18_2	R	1.153 ± 0.012	19.332	12.597 ± 0.091	1.05 ± 0.01 Ma	1.04 ± 0.01 Ma
CI HI 12_1	R	0.428 ± 0.004	8.135	4.581 ± 0.061	1.04 ± 0.02 Ma	
CI HI 12_2	R	0.428 ± 0.004	8.304	4.583 ± 0.060	1.04 ± 0.02 Ma	1.04 ± 0.02 Ma
CI HI 08_1	R	0.594 ± 0.006	9.392	6.479 ± 0.076	1.04 ± 0.02 Ma	
CI HI 08_2	R	0.594 ± 0.006	9.742	6.499 ± 0.074	1.05 ± 0.02 Ma	1.05 ± 0.02 Ma
CI HI 10_1	R	1.317 ± 0.013	19.088	15.221 ± 0.110	1.11 ± 0.01 Ma	
CI HI 10_2	R	1.317 ± 0.013	16.754	15.304 ± 0.119	1.11 ± 0.01 Ma	1.11 ± 0.01 Ma
CI HI 10_1	R	0.832 ± 0.008	11.498	9.571 ± 0.096	1.10 ± 0.02 Ma	
CI HI 10_2	R	0.832 ± 0.008	11.740	9.928 ± 0.098	1.14 ± 0.02 Ma	1.12 ± 0.02 Ma

Age calculations are based on the following decay (Steiger and Jäger, 1977) and abundance constants: $\lambda_{\beta}^{-} = 4.962 \times 10^{-10}$ a^{-1}; $\lambda_{\epsilon} = 0.581 \times 10^{-10}$ a^{-1}; $^{40}K/K = 1.167 \times 10^{-4}$ mol/mol. The ages for the standard minerals used for the calibration are for MMhb = 520.4 Ma, for LP-6 = 127.85 Ma and for HD-B1 = 24.07 Ma. Weights molten range between 1 and 2.5 g; subscripts 1, 2 = first and second argon determinations; P = polarity; N = normal polarity. R = reverse polarity.

the lower and middle parts of the El Golfo escarpment could not be directly related to R_1 magnetozone or to the Brunhes epoch and needed specific radiometric dating.

4.2.1. Age of magnetozone R_1

The oldest age, 1.12 ± 0.02 Ma, comes from the steep dipping lavas of El Tiñor volcanic edifice (R_1 magnetozone) near Puerto de La Estaca. A similar

age of 1.11 ± 0.01 Ma was obtained for the same formation 2.5 km apart (barranco. del Balõn). The latest emission vents of El Tiñor volcanic edifice corresponding to the R_1 magnetozone, the Ventejis volcanic group, gave two ages of 1.05 ± 0.02 and 1.04 ± 0.02 Ma (Table 3).

4.2.2. Age of magnetozones N_1 and R_2

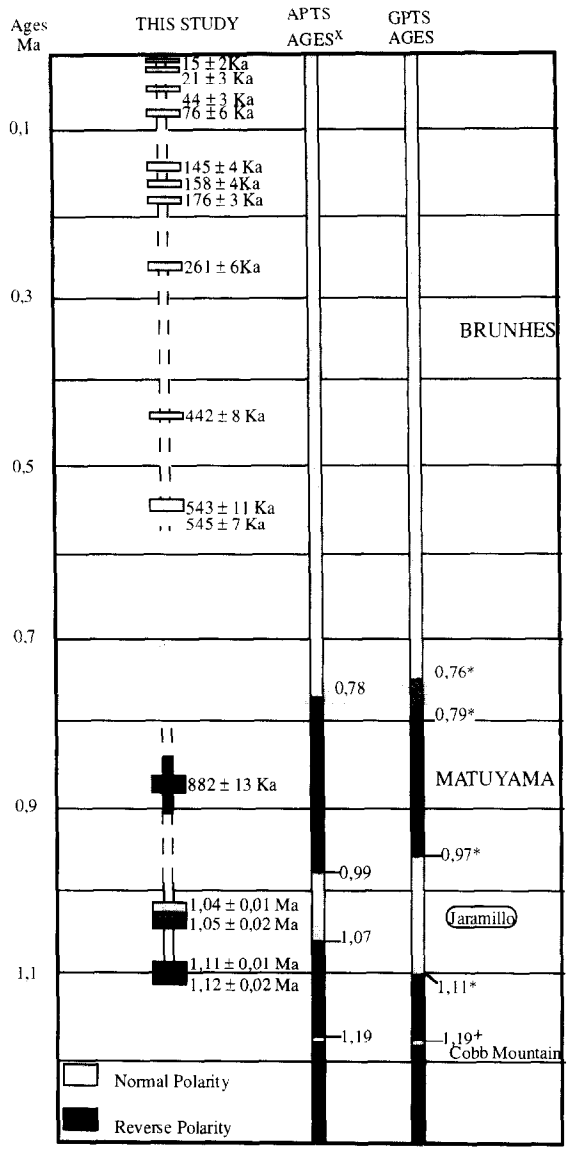
The subhorizontal lavas of the San Andres plateau, corresponding to N_1 magnetozone, yield an age of 1.04 ± 0.01 Ma. The late xenoliths-rich lavas from El Tiñor edifice (R_2 magnetozone) emission gave the youngest age: 882 ± 13 ka.

The El Tiñor edifice appears to have been active for a period of at least 0.20–0.27 m.y., taking into account the uncertainties on the ages. However, the main part of the edifice, the steep dipping lavas, seem to have been emitted in a surprisingly short period of about 100 ka.

4.2.3. Age of magnetozone N_2

Magnetic polarity profiles carried out in the 1100 m vertical escarpment of El Golfo identified only normal polarity volcanics. Before dating it was impossible to determine if one or several normal polarity intervals were present in this sequence. It was, however, evident that the uppermost lavas were contemporaneous with the Brunhes lavas overlying R_2 magnetozone in the northwestern part of the island. The previously published K-Ar ages for the El Golfo escarpment (Abdel-monem et al., 1972; Fúster et al., 1993) are inconsistent with the new data. A clear intraformational discordance separates the lower part of the El Golfo escarpment, where pyroclastic material densely intruded by dykes appears to be predominant, from the upper part, mainly formed by lava flows with fewer dykes. The upper volcanic suite ends with lava flows of transitional and trachytic compositions. This upper sequence grades without apparent discordance to the recent (Brunhes) rift lavas that cover most of the island. The ages obtained at the El Golfo escarpment were coherent with their stratigraphic position. The lowest sample (85 m a.s.l.) gave an age of 545 ± 11 ka, similar to the one obtained at the end of the gallery of Frontera, 543 ± 7 ka. One of the top lavas (295 m a.s.l.) below the intraformational discordance has an age of 442 ± 8 ka, whereas a lava flow overlying this unconformity (505 m a.s.l.) was dated at 261 ± 6 ka. The differentiated rocks (benmoreïtes and trachytes), probably the late stage of activity of El Golfo volcanic edifice, have an age of 176 ± 3 ka. The minimum duration

Table 3
Correlation of the polarity time scale determined from lavas of the island of El Hierro (this study) with the GPTS and APTS established by other authors



+ = $^{39}Ar/^{40}Ar$ age, Turrin et al., 1994; * = $^{39}Ar/^{40}Ar$ ages, Izett and Obradovich, 1994; [X] = Hilgen, 1991.

for the construction of El Golfo volcanic edifice may therefore be estimated to lie between 377 and 357 ka, taking into account the errors on the ages. Average lava accumulation rate for El Golfo edifice was thus about 1.4 mm/yr.

The first rift-related basaltic flow appears at 620 m a.s.l. and is 158 ± 4 ka. No significant gap in the volcanic activity is evident between the final stages of building of the El Golfo volcanic edifice indicated by the trachytic differentiates and the basaltic fissure eruptions at the rifts. The collapse that originated the El Golfo embayment occurred after accumulation of about 100 m of these lavas, coming from the northeastern rift. The distinction between lavas from El Golfo and the northeastern rift volcanic edifices is based in the opposing dip of both formations. The El Golfo lavas dip toward the northeast at the El Golfo escarpment, but their general structure is periclinal, pointing to a volcanic edifice centred in the present-day El Golfo embayment; the trachytic differentiates may indicate late stage dome emissions. The basaltic fissure lavas dip toward the sea in a SW trend, pointing to emission vents located along the northeastern rift.

The ages corresponding to lavas which originated from the present triple rift system range from 158 ± 4 to 11 ± 7 ka. Therefore only one polarity subchron corresponding to the Brunhes epoch post-dates the R_2 magnetozone.

5. Discussion

5.1. Geochronology and volcanic history

The complete magnetic stratigraphy of El Hierro is summarised in the sections of Fig. 6. The emergent part of the island has been built in about 1.2 m.y., a period corresponding to the upper part of the Matuyama reverse polarity and the Brunhes normal polarity chrons. The Jaramillo subchron is also present and helps to authenticate the geomagnetic polarity mapping.

The oldest R_1 magnetozone (1.12 ± 0.02 to 1.04 ± 0.02 Ma), recorded in the steep dipping lavas of the El Tiñor volcanic edifice, therefore correlates with the Matuyama pre-Jaramillo reverse polarity subchron. The N_1 magnetozone, corresponding to the subhorizontal lavas of the San Andrés plateau, have an age of 1.04 ± 0.0.1 Ma, consistent with the base

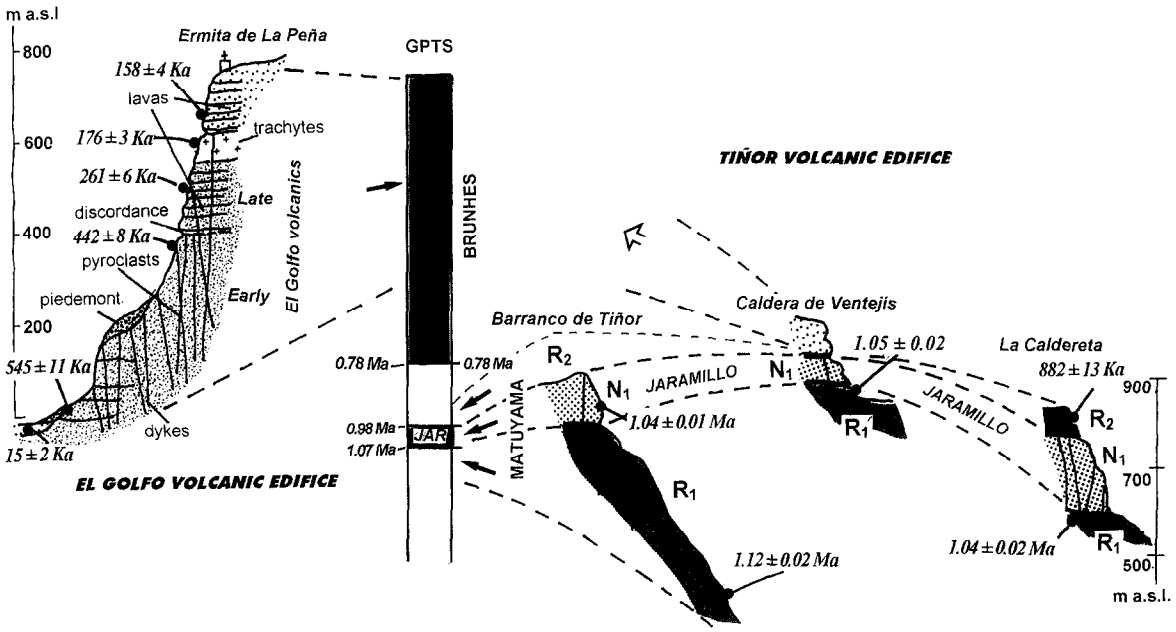


Fig. 6. Cross sections showing the correlation of magnetozones with the GPTS on El Hierro. See further explanation in the text.

of the Jaramillo normal polarity subchron. The lower limit of the Jaramillo event is constrained by two ages of 1.04 ± 0.02 Ma and one of 1.05 ± 0.02 Ma, close to the value (1.07 Ma) deduced from the Astronomic Polarity Time Scale (APTS) (Hilgen, 1991). The R_2 magnetozone comprising the late xenolith-rich lavas from El Tiñor edifice gave the youngest age: 882 ± 13 ka, compatible with the Matuyama post-Jaramillo reverse polarity subchron. The N_2 magnetozone, with ages ranging from 545 ± 11 ka to present occurs in the lavas cropping out in the El Golfo cliff, the upper part of Las Playas scarp and the formations that cap most of the surface of the island. This magnetozone correlates with the middle and upper part of the Brunhes chron.

The volcanic history of El Hierro is summarised by two consecutive main basaltic volcanic edifices: the El Tiñor edifice, encompassing the polarity changes within the upper part of the Matuyama chron, and the El Golfo edifice, constructed during the Brunhes chron. After a major tectonic event, between 882 ± 13 and 545 ± 11 ka (the volcanic and tectonic history of El Hierro is discussed in a paper in preparation) the El Golfo edifice was constructed attached to the western flank of the remains of the Tiñor edifice. Renewed activity began at the younger part of the Brunhes chron through a well developed triple-rift system, building on top of the previous edifice that was again tectonically disrupted. The activity of the triple-rift system has continued until present, covering the island with recent vents and lavas that totally or partially fill the tectonic scars.

5.2. Correlation of the magnetic polarity time scale of El Hierro to GPTS and APTS

Two independently calibrated polarity time scales are available. The first one (GPTS) is calibrated using conventional radiometric datings (K-Ar and ^{40}Ar-^{39}Ar methods). The second one is astronomically calibrated (APTS) using solar insolation periodicities observed in seawater O-isotopic curve. This scale has been developed by Shackleton et al. (1990) and Hilgen (1991). In Table 3, a comparison between the two time scales is given.

The ^{40}Ar-^{39}Ar ages obtained for the Brunhes–Matuyama boundary (Baksi et al., 1992; Spell and McDougall, 1992; Izett and Obradovich, 1994) which are, respectively, 780 ± 11, 770 and 780 ± 10 ka, are in good agreement with the APTS ages (Shackleton et al., 1990; Bassinot et al., 1994). However, the Jaramillo APTS boundaries show some discrepancies with those from the GPTS: the Jaramillo subchron extends from 1.11 to 0.97 Ma (Izett and Obradovich, 1994) or from 1.01 to 0.92 Ma (Spell and McDougall, 1992), while the APTS boundaries for the Jaramillo are 1.07 and 0.99 Ma (Shackleton et al., 1990).

In our results, all the recent lavas corresponding to the rift volcanism and to the El Golfo volcanic edifice are normal polarity and ranges in age from 545 to 15 ± 2 ka. Whatever the considered time scale, APTS or GPTS, this time period is a normal polarity period (the Brunhes normal chron), except for some very short excursions, such as the "Laschamp" or the "Blake" events that have not been detected.

The youngest dated lava which is related to the Ventejis eruptive centre, is 884 ± 13 ka, and has a reverse polarity. The second one, which is 1.04 ± 0.01 Ma has a normal polarity. These ages restrict the upper Jaramillo boundary in agreement with the aforementioned time scales.

The four oldest samples, with ages ranging from 1.04 ± 0.02 to 1.12 ± 0.02 Ma show reverse polarities. These ages, very close to lower Jaramillo boundary, are consistent only with the APTS; they lessen the duration of the Jaramillo subchron proposed by Izett and Obradovich (1994) who extended this event to 1.11 Ma. The paleomagnetic and radiometric results obtained here constrain the lower limit of the Jaramillo event to an age of 1.04 ± 0.02 Ma, which is slightly younger than the APTS age (1.07).

6. Conclusions

The combination of paleomagnetic polarity mapping and precise K-Ar dating have defined two main phases of volcanic activity on the island of El Hierro, correlative to the upper part of the Matuyama reverse polarity and to the middle–upper part of the Brunhes normal polarity chrons. The Jaramillo subchron has also been detected, with a duration consistent with the APTS (Shackleton et al., 1990) but about 50% shorter than the GPTS defined by Izett and Obradovich (1994).

The intense and continued Quaternary eruptive activity shown by the development of El Hierro is in agreement with the assumption that this island is presently being built above the Canaries hotspot. The volcanic history of the island is characterised by three main stages, the oldest and intermediate ones corresponding to two consecutive, fast developed, high aspect ratio volcanic edifices. Overgrowth instability resulted in successive gravitational collapses that restored the island to a stable configuration. The third stage is still in progress.

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